DonProof PU-P150

Spray applied, pure polyurea membrane for waterproofing and protection

DESCRIPTION

DonProof PU-P150 is a sprayable, high-quality, two-component, 100% solids, aromatic pure polyurea coating that cures to form a solid, seamless, watertight and waterproof, high density polyurea membrane, with high mechanical properties.

DonProof PU-P150 is designed to have excellent adhesion on many common construction surface such as concrete, ceramic tiles, metals, aluminum, plywood, asphalt/ bituminous sheets when used with suitable primer.

DonProof PU-P150 adaptability and drying time makes it suitable for application on uneven surfaces and in areas of any shape, whether curved or squared.

APPLICATIONS

Waterproofing and protection of:

- Sloped and flat roofs (walkable), balconies, and pverhangs, with additional top coat for UV-protection for exposed areas.
- Under XPS insulation as part of an inverted or vegetative roof system.
- Retaining walls and foundations.
- Concrete/steel water storage tanks.
- Car park and bridge decks.
- Desalination and water treatment plants.
- Swimming pools, aquariums, lakes.
- Green roofs and planters.

ADVANTAGES

- Solvent free.
- Excellent mechanical properties, high tensile and tear strength, high abrasion resistance.
- Excellent adhesion to all common substrates.
- Fast application and fast curing/overcoating time.
- Strong, flexible and hard-wearing membrane.
- Seamless and monolithic.
- Excellent thermal resistance. Max service temperature 90ºC. Withstands short-duration temperatures up to 250ºC.
- Cold Resistance: down to minus 30ºC.
- Excellent stability and durability.
- Friendly to the environment, free from harmful VOC compounds.
- Optimum maintenance and cleaning properties for the cured surface.
- Resistant to root penetration.

TECHNICAL PROPERTIES @ 25ºC:

Colour:	Various
Density: (cured film)	1.10 ± 0.05 g/cm³
Solid content: ISO 1768	100%
Tack free time:	5 - 7 sec
Recoating time:	4 min - 3 hr
Light pedestrian traffic time:	12 hr
Vehicle traffic time:	2 - 3 days
Service temperature:	-30 to 90ºC
Shore hardness: ASTM D2240 Shore A @ 7 days	≥ 75 ± 5
Taber abrasion: ASTM D4060 (1000 g, 1000 cycle) CS17 wheel	20 milligram
Tensile strength: ASTM D412	≥ 15 MPa
Bond strength to concrete:	≥ 4 MPa
Bond strength to steel*:	≥ 2.0 MPa
Elongation: ASTM D412	≥ 300%
Tear strength: ASTM D624	≥ 80 N/mm
Crack Bridging: ASTM C1305 @ 23ºC	no cracks occurred after 10 cycles
Water vapor transmission: ASTM E96	Medium 0.6 to 6.0 g/(m²/hr)
Water permeability: Din 1048 @ 5 bars	Nil
Resistance to roots: PD CEN/TS 14416:2014	No penetration

**When primed with DonProof Primer PU at 3 - 4 hr recoating time.*

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METHOD OF USE

SURFACE PREPARATION

The surface should be clean, dry, sound, and free from oil, grease, and wax contamination. Cement laitance, loose particles, mold release agents, or curing membranes must be removed.

Fill surfaces irregularities with a suitable epoxy resin mortar. The maximum atmospheric relative humidity should not exceed 85%. New concrete structures need to dry for at least 28 days. Do not proceed with the application in the presence of moisture.

Concrete surfaces should achieve an open pore surface (surface Preparation index -CSP- 3 to 6) by grit blasting, milling, or sandblasting.

Substrate humidity or the presence of water on the substrate can affect the membrane's adhesion, which will have a negative effect on the result of the system. Working at ambience temperatures below 5ºC or more than 50ºC is not recommended. The substrate's temperature must be at least 3ºC above the measured due point temperature if any.

PRIMING

It is recommended to prime concrete substrates using DonProof Primer EP epoxy primer and leave it to cure for 12 - 24 hrs @ 25ºC.

Steel and galvanized steel substrates should be primed using DonProof Primer PU. Refer to the Technical Datasheet of the primers for further information.

Contact DCP Technical Department for best priming options for non-porous substrates such as steel and ceramic tiles, as well as for other substrate types based on project conditions.

APPLICATION

DonProof PU-P150 is hot spray applied; a suitable two components high-pressure heated spray machine (2.500 - 3.000 psi), with a heater temperature of 70 - 75ºC and a hose temperature of 70ºC should be used. Both components must be heated up to approximately 75ºC. (consult DCP Technical Department for further details).

	Component A (Iso)	**Component B (Resin)**
Colour:	Clear brown	Coloured as per request
Density: ISO 1675	1.10 ± 0.01 g/cm³	1.07 ± 0.01 g/cm³
Viscosity @ 30ºC:	400 - 800 cps (N_2S_{50})	400 - 800 cps (N_2S_{50})
Mix ratio:		
By weight	1.03	1.00
By volume	1	1

Stir/mix component B (resin) well before use using a drum stirrer to homogenize the pigment and disperse any settlement. Failing to do so may cause color variances, foaming, sticky coating, and negatively affect DonProof PU-P150 performance.

Insert the pumps into the drums and attach the hoses to the equipment, set up the hoses and machine temperatures and start the equipment following the manufacturer instructions.

Apply the material using the spray gun. always apply DonProof PU-P150 perpendicularly to the substrate regardless of whether the surface is horizontal or vertical. This is extremely important, otherwise, it will cause gaps in the layer and incomplete seal.

Apply crossed coat continuously as needed to achieve the desired thickness according to the final use. When utilising a cross-hatch spray pattern, the minimum suggested application thickness for DonProof PU-P150 is typically 1.4 mm. After being applied, the material is suitable for light foot traffic) after about 12 hours and is completely functional after about 2 - 3 days.

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CONSUMPTION

Consumption will depend on the type of application, weather conditions, or substrates' nature.

» Recommended consumption is approximately: 1.6 kg/m² to provide 1.5 mm dry film thickness.

UV PROTECTION COAT

As with all aromatic coatings, DonProof PU-P150 should be sealed with a suitable polyurethane protection coat to prevent yellowing due to UV rays exposure. However, this will not cause any negative effect on the physical properties of the product.

Consult DCP Technical Department for specific recommendations for an aliphatic topcoat to guarantee its durability.

CLEANING

All tools should be cleaned immediately with a suitable solvent and diluent, refer to the spray machine manufacturer for specific recommendations depending on the machine used.

PACKAGING

DonProof PU-P150 is available in (component A (Iso): 221 kg drums) and (component B (Resin): 214 kg drums).

STORAGE

DonProof PU-P150 has a shelf life of 12 months from date of manufacture if stored in a dry place, away from direct sunlight, extreme heat, cold, or moisture and in its original unopened pails at temperatures between 5°C and 30°C. Higher storage temperatures may reduce shelf life of the product.

Once the tin has been opened, the product must be used. Once opening, component B drum must be agitated mechanically before inserting the transfer pumps and use.

If these conditions are exceeded, contact DCP Technical Department for advice.

CAUTIONS

HEALTH AND SAFETY

Apply in well ventilated areas. Do not smoke. Do not apply near naked flames. In closed areas use force ventilation and carbon active masks.

For further information, refer to the Material Safety Data Sheet.

FIRE

DonProof PU-P150 is nonflammable

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A wide range of construction chemical products are manufactured by DCP which include:

- Concrete admixtures.
- Surface treatments
- Grouts and anchors.
- Concrete repair.
- Flooring systems.
- Protective coatings.
- Sealants.
- Waterproofing.
- Adhesives.
- Tile adhesives and grouts.
- Building products.
- Structural strengthening.

Note:
We endeavour to ensure that any information, advice or recommendation we may give in product literature is accurate and correct. However, because we have no control over where and how products are applied, we cannot accept any liability arising from the use of the products.